

BS-CH101 / BS-CH201 — Chemistry-I

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Module 3 Exam Notes: Intermolecular Forces and Potential Energy Surfaces

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Intermolecular Forces and Potential Energy Surfaces

Intermolecular forces are attractive or repulsive forces between molecules or ions. They are weaker than covalent bonds but strongly affect boiling point, solubility, viscosity, and real-gas behaviour.

1.1 Ionic, Dipole and Van der Waals Interactions

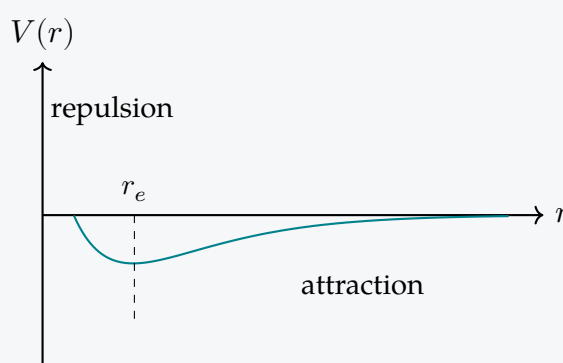
Ionic interactions are strongest among common non-covalent interactions. Dipole-dipole forces occur between polar molecules. London dispersion forces arise from temporary induced dipoles and exist in all molecules.

Interaction	Origin	Example
Ion-ion	Attraction between full charges	Na^+ and Cl^-
Ion-dipole	Ion attracts polar molecule	Hydration of Na^+
Dipole-dipole	Permanent dipoles align	HCl molecules
London dispersion	Instantaneous induced dipoles	Noble gases, hydrocarbons
Hydrogen bonding	H attached to F, O, or N interacts with lone pair	Water, alcohols

1.2 Potential Energy Curve

The potential energy between two particles depends on separation distance. At equilibrium distance r_e , attractive and repulsive forces balance and potential energy is minimum.

Potential Energy Curve for Intermolecular Interaction



1.3 Equations of State for Real Gases

Ideal gas equation assumes no intermolecular force and zero molecular volume. Real gases deviate at high pressure and low temperature.

Van der Waals Equation

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

where a corrects attractive forces and b corrects finite molecular volume.

1.4 Critical Phenomena

The critical temperature T_c is the temperature above which a gas cannot be liquefied by pressure alone. At the critical point, liquid and vapour become indistinguishable.

Critical Constants for a Van der Waals Gas

$$V_c = 3nb, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

Exam Points

- Compare intermolecular forces by strength and origin.
- Draw the potential energy curve and mark r_e and minimum energy.
- For real gases, explain physical meaning of constants a and b .
- Define critical temperature, pressure, and volume clearly.

2. Detailed Notes on Forces, Real Gases and Critical Behaviour**2.1 Origin and Strength of Intermolecular Forces**

Intermolecular force strength depends on charge, polarity, polarizability, and molecular shape. Ionic interactions are strong because full charges are involved. Dispersion forces are weak individually but become important for large molecules with many electrons.

Force	Approximate strength trend	Property affected
Ion-ion	Very strong	Melting point of ionic solids
Ion-dipole	Strong	Hydration and solubility of salts
Hydrogen bond	Moderate to strong	Boiling point of water and alcohols
Dipole-dipole	Moderate	Boiling point of polar molecules
London dispersion	Weak to moderate	Liquefaction of non-polar gases

2.2 Potential Energy and Force Relation

The force between two particles is related to the potential energy curve by:

$$F = -\frac{dV}{dr}$$

At the equilibrium separation r_e , potential energy is minimum, so:

$$\left(\frac{dV}{dr}\right)_{r=r_e} = 0, \quad F = 0$$

For $r < r_e$, repulsive force dominates. For $r > r_e$, attractive force dominates.

Exam Interpretation of Potential Energy Curve

If two molecules are very far apart, $V(r) \approx 0$. As they approach, attraction lowers the potential energy. At r_e , the most stable separation is reached. If they come closer than r_e , electron-cloud repulsion sharply increases energy.

2.3 Lennard-Jones Potential

A common model for non-bonded interaction is:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$

The r^{-12} term represents short-range repulsion and the r^{-6} term represents London attraction. ϵ indicates well depth, i.e. strength of attraction.

Term	Physical meaning
$(\sigma/r)^{12}$	Repulsion due to electron-cloud overlap
$-(\sigma/r)^6$	Attractive dispersion interaction
ϵ	Depth of potential well; higher value means stronger attraction
r_e	Separation at minimum potential energy

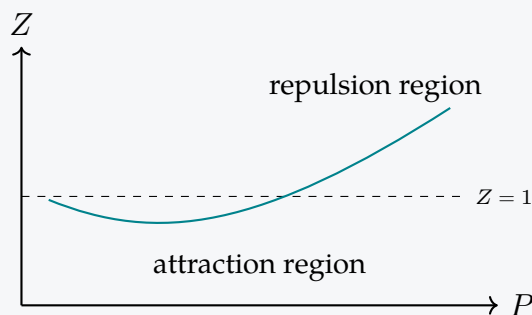
2.4 Compressibility Factor and Deviation from Ideality

Deviation from ideal gas behaviour is often measured using compressibility factor:

$$Z = \frac{PV}{nRT}$$

For an ideal gas, $Z = 1$. For a real gas, $Z < 1$ indicates attraction-dominated behaviour and $Z > 1$ indicates repulsion or finite-volume dominated behaviour.

Compressibility Factor: Qualitative Behaviour



2.5 Critical Constants from Van der Waals Equation

For one mole:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

At the critical point, the isotherm has a point of inflection:

$$\left(\frac{\partial P}{\partial V_m}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial V_m^2}\right)_T = 0$$

Solving these conditions gives:

$$V_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

These relations are often asked as short derivation or formula-based questions.

Solved Example: Critical Volume

If van der Waals constant $b = 0.040 \text{ L mol}^{-1}$, then:

$$V_c = 3b = 3(0.040) = 0.120 \text{ L mol}^{-1}$$

How to Score in This Module

- Write force origin, example, and property affected for each intermolecular force.
- In potential energy questions, always mark attraction, repulsion, r_e , and energy minimum.
- For real gases, explain that a corrects pressure and b corrects volume.
- Use $Z = PV/nRT$ to discuss positive and negative deviation.

2.6 Hydrogen Bonding in Detail

Hydrogen bonding occurs when hydrogen is covalently bonded to a highly electronegative atom such as F, O, or N and interacts with a lone pair on another electronegative atom.

Type	Meaning	Example
Intermolecular hydrogen bonding	Hydrogen bond between different molecules	Water, alcohols, HF
Intramolecular hydrogen bonding	Hydrogen bond within the same molecule	o-nitrophenol type structures
Strong hydrogen bonding	Shorter and more directional interaction	HF, water network
Weak hydrogen bonding	Longer and less directional interaction	C-H based weak interactions

Exam Explanation: Boiling Point of Water

Water has a much higher boiling point than H_2S because oxygen is highly electronegative and forms extensive intermolecular hydrogen bonding. More heat is

required to break this associated network before molecules escape into vapour phase.

2.7 Corresponding States and Reduced Variables

The law of corresponding states states that gases behave similarly when compared at the same reduced temperature and reduced pressure.

Reduced Variables

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}, \quad V_r = \frac{V}{V_c}$$

where T_c , P_c , and V_c are critical constants.

2.8 Boyle Temperature

At Boyle temperature, a real gas behaves nearly ideally over a considerable pressure range because attractive and repulsive deviations approximately cancel.

$$Z \approx 1$$

For a van der Waals gas, the Boyle temperature is approximately:

$$T_B = \frac{a}{Rb}$$

Solved Example: Boyle Temperature

If $a = 1.36 \text{ L}^2 \text{ atm mol}^{-2}$ and $b = 0.0318 \text{ L mol}^{-1}$:

$$T_B = \frac{a}{Rb} = \frac{1.36}{(0.0821)(0.0318)}$$

$$T_B \approx 521 \text{ K}$$

2.9 Joule-Thomson Effect

When a real gas expands through a porous plug from high pressure to low pressure without external work and without heat exchange, its temperature may change. This is called Joule-Thomson effect.

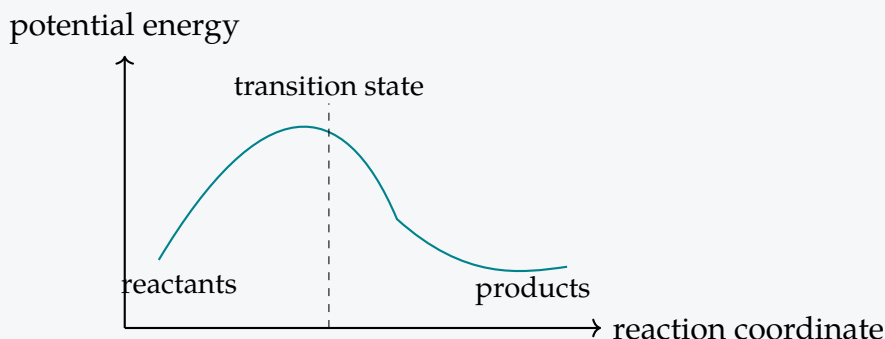
Inversion temperature: The temperature at which Joule-Thomson coefficient becomes zero. Below it, gas cools on expansion; above it, gas warms on expansion.

Condition	Joule-Thomson coefficient	Observation
$T < T_i$	Positive	Gas cools on expansion
$T = T_i$	Zero	No temperature change
$T > T_i$	Negative	Gas warms on expansion

2.10 Potential Energy Surface in Chemical Reactions

A potential energy surface shows how molecular energy changes with geometry. Reactants, products, intermediates, and transition states are represented as regions on the surface.

Reaction Path on a Potential Energy Surface



Previous-Year Style Questions

1. Compare ion-dipole, dipole-dipole, hydrogen bonding and dispersion forces.
2. Explain the potential energy curve for two interacting molecules.
3. Derive or state critical constants of a van der Waals gas.
4. Explain compressibility factor and deviation from ideal behaviour.
5. Write short notes on Joule-Thomson effect and inversion temperature.

2.11 Applications of Intermolecular Forces in Engineering Chemistry

Intermolecular forces are not only theoretical; they explain important material and chemical behaviour.

Application	Force involved	Importance
Solubility of salts in water	Ion-dipole force	Hydration stabilizes ions in solution
Polymer strength	Dipole interaction and hydrogen bonding	Controls toughness, flexibility and melting range
Adsorption on catalyst	van der Waals and polar interactions	Reactants attach to active surface sites
Boiling point of liquids	Hydrogen bonding and dispersion	Stronger forces increase boiling point
Protein structure	Hydrogen bonding and hydrophobic interaction	Maintains secondary and tertiary structure

Exam Answer: Like Dissolves Like

Polar solvents dissolve ionic and polar solutes because ion-dipole or dipole-dipole interactions compensate for the energy needed to separate solute particles. Non-polar solutes dissolve better in non-polar solvents because dispersion forces are similar in

both.

Quick Revision — Module 3

Intermolecular Forces and Potential Energy Surfaces | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$F = -\frac{dV}{dr}$	Force is negative gradient of potential energy.
$PV = nRT$	Ideal gas equation.
$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$	Van der Waals real-gas equation.
$V_c = 3nb$	Critical volume for van der Waals gas.
$P_c = \frac{a}{27b^2}$	Critical pressure for one mole form.
$T_c = \frac{8a}{27Rb}$	Critical temperature for one mole form.

One-Line Definitions

- **Intermolecular force:** Force acting between molecules or ions.
- **Dipole moment:** Product of charge and separation in a polar molecule.
- **Hydrogen bond:** Strong dipole interaction involving H bonded to F, O, or N.
- **Potential energy surface:** Energy variation with molecular geometry.
- **Real gas:** Gas that deviates from ideal behaviour due to volume and attraction.
- **Critical temperature:** Highest temperature at which gas can be liquefied by pressure.

Frequently Asked / Viva Questions

Q1. What are intermolecular forces?

Forces of attraction or repulsion acting between separate molecules or ions.

Q2. Arrange common forces in increasing strength.

London dispersion < dipole-dipole < hydrogen bonding < ion-dipole < ion-ion.

Q3. Why does boiling point increase with stronger intermolecular force?

More energy is needed to separate molecules into vapour.

Q4. What is equilibrium separation in a potential energy curve?

It is the distance where potential energy is minimum and net force is zero.

Q5. What causes repulsion at very small distance?

Overlap of electron clouds and nuclear repulsion.

Frequently Asked / Viva Questions (contd.)

Q6. Why do real gases deviate from ideal behaviour?

Molecules have finite volume and attractive forces.

Q7. What is the role of a in van der Waals equation?

It corrects pressure for intermolecular attraction.

Q8. What is the role of b in van der Waals equation?

It corrects volume for finite molecular size.

Q9. Define critical point.

The point at which liquid and vapour phases become indistinguishable.

Q10. What is critical temperature?

The highest temperature above which gas cannot be liquefied by pressure alone.

Summary Reference Table

Concept	Meaning	Must write in exam
Intermolecular forces	Non-covalent forces between species	Strength order and examples
Potential curve	Energy versus separation	Minimum at r_e
Real gas	Non-ideal gas	Van der Waals correction terms
Critical point	End point of liquid-vapour boundary	T_c , P_c , V_c definitions